

Fourth-Order Perturbations of Pure-Weyl Black Holes

Exact Ricci–Bach Composition, Horizon Pairing, and Endpoint Nonselection

GPT-5.6.sol*

Claude Fable 5†

Working draft, 23 July 2026

Abstract

We ask whether familiar one-ended Lorentzian endpoint conditions isolate the Einstein sector of four-dimensional pure Weyl gravity around a black hole. We first classify the Bach-flat locus in a declared static Laurent ansatz, recover the Mannheim–Kazanas component, and derive a residual-basic horizon generator. On its nondegenerate static domain, the associated Lee–Wald Hamiltonian, Wald entropy, and signed surface-gravity temperature obey an exact first law at every simple horizon. We correct an important sheet distinction: on the complete Laurent locus the metric is Einstein precisely when $\gamma = 0$ and $w = 1$, whereas $\gamma = 0$ alone suffices only on the Mannheim–Kazanas component through $w = 1$.

On Schwarzschild, linearizing the Bach tensor gives an exact Ricci-carrier composition. It defines an exact sequence from Bach solutions to curvature solutions, not a canonical Einstein/additional direct sum. In the axial $\ell = 2$ carrier, $\psi_{ab} = \delta R_{ab}$ obeys a second-order Lichnerowicz-type equation. For nonzero real frequency its horizon system has a two-dimensional analytic ingoing ψ -space, so future-horizon regularity does not force $\psi = 0$. The action-derived Lee–Wald current makes the Einstein self-block isotropic for conjugate Regge–Wheeler waves. Controlled finite-series fixtures at three frequencies exhibit nonzero Einstein–additional cross pairing and nonzero additional-sector pairing; these fixture results are not promoted to a symbolic all-frequency theorem.

At infinity the earlier $\ell = 2$, $M_{\text{BH}}\omega = 3/5$ claim that finite radial pairing selects exactly the Einstein sector is superseded. A Phase-3 audit found that the axial metric reconstruction had also omitted the independent $v\varphi$ Ricci equation. On the declared $\ell = 2$, $M_{\text{BH}} = 1$, $\omega \in [1/2, 3/4]$ pilot this equation propagates as $C' = -2C/r$; its zero fibre is an exact six-dimensional formal module. The repaired non-Einstein X_0 direction is radially finite in the fixed representative and under the true rate-zero Einstein shear, but its cross pairing with the true oscillatory Einstein shear diverges. Thus no representative-independent axial selection theorem follows. Independently, the polar restriction-stable module contains a mixed Einstein/additional line that is finite and generically nonradical, with an explicit algebraic exceptional locus. Thus neither leading endpoint diagnostics nor finite formal radial pairing isolates the Einstein image. Separately, on the same axial pilot, the exact three-dimensional L^2 wave-packet trace spaces at $\mathcal{I}^-$ and $\mathcal{I}^+$ carry action-derived endpoint flux Grams of rank three, zero radical, and inertia $(1, 2, 0)$ for $\alpha_W > 0$, with no frequency wall on the pilot interval. An exact differential-module reduction further gives two spin-two Regge–Wheeler factors and one spin-one Regge–Wheeler factor. Their real-potential Wronskians prove that the global map from future-horizon-regular data to the incoming $\mathcal{I}^-$ trace is invertible for every real $\omega > 0$. Hence the complete indefinite incoming endpoint space is globally populated; its spin-two extension is hyperbolic and its orthogonal spin-one quotient

*OpenAI model and principal programme author.

†Anthropic model and computational coauthor. The project was commissioned and coordinated by Asger Alstrup Palm (asger@area9.dk), who initiated the questions, exercised editorial control, and serves as corresponding human contact, but claims no technical authorship.

line is negative. The outgoing reflection block, non-real poles, a complete asymptotically flat scattering operator, full time-domain stability, damped quasinormal spectrum, Hawking state, and quantum unitarity remain open. The exact filtration does, however, exclude exponentially growing axial $\ell = 2$ separated modes. The local Cauchy restriction $\psi|_{\Sigma} = \nabla_n \psi|_{\Sigma} = 0$ still selects the Einstein kernel at the linear level.

AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every certified statement below is bound to a machine-readable certificate and a separately implemented verifier. The cross-flux result uses controlled high-precision series evaluation with exact recurrence data and independent frequencies; it is reported as fixture-level evidence, not interval-certified nonvanishing. All local tensor identities and polynomial classifications are exact. The manuscript remains conditional on final human review.

Contents

1	The physical question	3
1.1	Claims and dependency tags	3
2	Relation to existing branch-selection programmes	3
3	Pure Weyl gravity and its covariant current	4
4	Static spherical Bach-flat family	4
5	Residual-basic energy and the static first law	5
6	Ricci–Bach composition and the axial exact sequence	6
7	The Ricci carrier reaches the future horizon	7
8	Action-derived horizon flux	8
9	Asymptotic symbol and endpoint nonselection	9
9.1	Complete axial reconstruction on the $\ell = 2$ pilot	9
9.2	Exact axial null-endpoint wave-packet flux	10
9.3	Polar mixed finite line and its exceptional wall	17
9.4	Local Cauchy selection remains available	18
10	Polar chain: composition, reach, current, and disposition	18
11	Horizon monodromy temperature	20
12	Result ledger	22
13	What is not proved	23

14 Certificate and reproduction ledger	23
15 Outlook	25
A Explicit axial endpoint Grams and uniform bounds	25

1 The physical question

Fourth-order gravitational equations contain more local solutions than the Einstein equations. A common proposal is to retain the Einstein branch by a boundary condition. Around a one-ended Lorentzian black hole, a first test is:

Do future-horizon analyticity and the same leading outer admissibility tests used for Einstein waves force the additional Ricci carrier to vanish?

The answer in the certified Schwarzschild axial $\ell = 2$ mode laboratory is no: the curvature carrier has regular ingoing horizon data, the leading real-frequency symbol does not separate it from the Einstein carrier, and controlled fixtures have nonzero action-derived horizon pairing. This is an endpoint-admissibility result, not a classification of all local differential boundary conditions. In particular, setting the Ricci carrier and its normal derivative to zero on a Cauchy surface is a local linear Einstein-sector restriction. Whether that restriction is natural, interaction-stable, or compatible with a physical asymptotic phase space is open.

1.1 Claims and dependency tags

The static family and local operator identities carry LOCAL-ALGEBRAIC. Harmonic branch, horizon, current, and asymptotic statements also carry REDUCED-MODE. No result in this manuscript is promoted to LORENTZIAN-CAUSAL: there is no complete exterior Green-hyperbolicity theorem, metric-falloff phase space, or classified set of local differential boundary operators.

2 Relation to existing branch-selection programmes

The static Bach-flat family itself is classical [2, 3]. Conformal-gravity thermodynamics with additional spin-two hair has also been studied in asymptotically AdS and Einstein–Weyl settings [14]. Our static contribution is therefore not a new family: it is the exact residual-basic normalization and simultaneous signed first-law identity within a declared local quotient. A physical asymptotic normalization has not yet been selected. In particular, Lü, Pang, Pope, and Vázquez-Poritz organize AdS thermodynamics with an independent massive-spin-two-hair conjugate pair. Our one-term identity instead follows after quotient-basic normalization on a restricted static parameter family. The different boundary ensemble, generator normalization, and independent variables mean that the two first laws are complementary, not contradictory.

Maldacena’s Einstein-from-conformal construction selects an Einstein sector by Euclidean AdS or late-time de Sitter boundary data [8]. Critical- and logarithmic-gravity analyses similarly separate massless, massive, and generalized modes by asymptotic conditions [9, 13, 10, 11, 12]. The null Einstein self-pairing and nonzero cross pairing found below should be read in that established degenerate-operator context, rather than as the first appearance of null massless modes paired with generalized solutions. Our different question is whether one-ended Lorentzian Schwarzschild endpoint tests force the Ricci carrier to vanish.

Using δR_{ab} to expose higher-derivative spin-two modes is also a known strategy in fourth-order gravity [15]. Previous Weyl-black-hole quasinormal analyses often deliberately obtained a second-order master equation by conformal transport rather than retaining the full fourth-order Bach solution space [16]. Effective-Einstein perturbations of conformally related nonsingular metrics [17] and scalar or electromagnetic test-field spectra [18] are different problems: test-field stability is not stability of the dynamical fourth-order Bach sector. Contemporary quadratic-gravity ringdown work finds both ordinary and additional spin-two sectors [19], while time-domain quadratic-gravity studies track massive-sector tails [20]. The prospective advance here is narrower: the exact pure-Weyl Ricci–Bach sequence, its action-derived Schwarzschild horizon pairing, and failure of the tested endpoint conditions to imply $\delta R_{ab} = 0$. The first-order-system analysis follows the general modified-gravity asymptotic methodology of [21]; in particular, that methodology makes the unresolved repeated-root Jordan audit an essential next step rather than a technical detail. Precise decay of the metric coefficients must ultimately be compared with the Schwarzschild Einstein benchmark [22], not inferred from bounded curvature carrier variables alone.

3 Pure Weyl gravity and its covariant current

We use signature $(-+++)$ and action

$$S_W[g] = \alpha \int_M \sqrt{-g} C_{abcd} C^{abcd} d^4x. \quad (1)$$

The field equation is the Bach equation $B_{ab} = 0$, with convention

$$B_{ab} = \nabla^c \nabla^d C_{acbd} + \frac{1}{2} R^{cd} C_{acbd}. \quad (2)$$

The covariant presymplectic current is obtained from the literal variation of the curvature-momentum potential of this action. Its normalization is fixed by the off-shell Green identity

$$\partial_t F^t(h_A, h_B) + \partial_r F^r(h_A, h_B) = 4\alpha \int_{S^2} \sqrt{q} (h_B^{ab} \delta B_{ab}[h_A] - h_A^{ab} \delta B_{ab}[h_B]). \quad (3)$$

Thus the current is conserved for pairs of linearized Bach solutions and is not a freely rescaled master-equation Wronskian.

4 Static spherical Bach-flat family

In Schwarzschild gauge,

$$ds^2 = -B(r)dt^2 + B(r)^{-1}dr^2 + r^2 d\Omega_2^2, \quad (4)$$

the declared Laurent ansatz is

$$B(r) = w - \frac{u}{r} + \gamma r - kr^2 + \frac{c_2}{r^2} + c_3 r^3. \quad (5)$$

Theorem 4.1 (Complete Bach-flat locus in the Laurent ansatz). *The metric is Bach-flat if and only if*

$$c_2 = c_3 = 0, \quad w^2 + 3u\gamma = 1. \quad (6)$$

The Mannheim–Kazanas parametrization is recovered by

$$w = 1 - 3\beta\gamma, \quad u = \beta(2 - 3\beta\gamma).$$

On the complete Laurent locus the metric is Einstein if and only if

$$\gamma = 0, \quad w = 1. \quad (7)$$

On the Mannheim–Kazanas component through $w = 1$, this reduces to the single condition $\gamma = 0$.

Completeness here refers only to the Laurent ansatz. The certificate does not claim a new proof of the global static spherical classification. The sheet qualification in (7) follows already from the scalar curvature

$$R = 12k - \frac{6\gamma}{r} + \frac{2(1-w)}{r^2}. \quad (8)$$

In particular, the allowed point $(\gamma, w) = (0, -1)$ is not Einstein. The original static certificate’s phrase “Einstein iff $\gamma = 0$ ” is therefore used only on its explicitly parametrized Mannheim–Kazanas sheet; the manuscript does not propagate it to the complementary Laurent sheet.

The form-preserving local $\text{Diff} \times \text{Weyl}$ transformations act as dilation and translation on the cubic

$$Q(x) = -ux^3 + wx^2 + \gamma x - k, \quad x = r^{-1}.$$

Their orbit rank is two. A single continuous invariant may be taken as

$$J = u^2 \text{disc}(Q),$$

while $\text{disc}(Q) = 0$ is the degenerate-horizon locus. The conformal translation may fail globally where its Weyl factor vanishes; it is used only as a local orbit statement.

An exact non-Einstein three-horizon fixture is

$$(\beta, \gamma, k) = \left(\frac{3}{2}, \frac{12}{19}, \frac{1}{19} \right), \quad rB(r) = -\frac{1}{19}(r-1)(r-3)(r-8).$$

All curvature invariants are finite at the simple horizons. In ingoing Eddington–Finkelstein coordinates,

$$ds^2 = -B(r)dv^2 + 2dvdr + r^2 d\Omega_2^2,$$

the metric and Bach equation are regular there.

5 Residual-basic energy and the static first law

For the chart generator ∂_t , the bare static Lee–Wald one-form is not integrable. The obstruction disappears when the generator is normalized consistently with the residual quotient.

Theorem 5.1 (Normalized static generator). *On a connected component with $u \neq 0$, choose the oriented generator*

$$\chi = u \partial_t, \quad u = \beta(2 - 3\beta\gamma). \quad (9)$$

Residual basicness and dilation weight determine the general normalization as $uf(J)\partial_t$. We select $f(J) = 1$. Its Hamiltonian is

$$H = -16\pi\alpha\beta^2 D_2, \quad D_2 = 9\beta^2\gamma^2 k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k. \quad (10)$$

At every simple horizon r_h , the Wald entropy

$$S_h = 64\pi^2\alpha\beta \frac{2 - 3\beta\gamma + \gamma r_h}{r_h} \quad (11)$$

and normalized temperature

$$T_h = \frac{uB'(r_h)}{4\pi} \quad (12)$$

satisfy

$$dH = T_h dS_h \quad (13)$$

identically modulo $B(r_h) = 0$.

Here T_h is the *signed* surface-gravity temperature appropriate to the oriented first-law identity; the positive Hawking temperature would use $|uB'(r_h)|/(4\pi)$. On a component where $u < 0$, future-directedness uses $-\chi$, flipping both H and T_h and preserving the first law. The locus $u = 0$ is excluded because this normalization degenerates. For another allowed $f(J)$, one has $T_f = f(J)T_h$ and a reparametrized Hamiltonian satisfying $dH_f = f(J)dH$; the first law remains true. Thus residual basicness fixes a class, not a unique physical normalization in the absence of an asymptotic clock.

This is a theorem on the static parameter slice, not yet a general physical-process first law. Its linear dynamical extension nevertheless passes a strong gauge audit: arbitrary time-dependent spherical Weyl rescalings and diffeomorphisms have exactly zero charge and flux, the entropy is conformally invariant on the symbolic family, and the normalized generator extension is unique at linear order under the declared boundary scalar rule.

6 Ricci–Bach composition and the axial exact sequence

The parity-specific formulas are restrictions of one Ricci-flat identity.

Proposition 6.1 (Linearized Bach tensor on a Ricci-flat background). *For a perturbation of a four-dimensional Ricci-flat metric,*

$$\delta B_{ab} = \frac{1}{2}\square(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\square\delta R. \quad (14)$$

Proof. Rewrite the four-dimensional Bach tensor in Ricci variables using the contracted Bianchi identity, then retain the terms linear in Ricci curvature about $R_{ab} = 0$. Background derivatives and the curvature commutator give the displayed Weyl action. The repository verifies (14) independently, component by component, on the complete axial and polar $\ell = 2$ Regge–Wheeler-gauge carriers. That computer check is carrier-complete but is not presented as a machine proof for arbitrary harmonic content. $\square$

We now specialize the background to Schwarzschild and use axial Regge–Wheeler gauge,

$$h_{t\phi} = h_0(t, r)S_{\ell m}, \quad h_{r\phi} = h_1(t, r)S_{\ell m}.$$

The $\ell = 2$ linearized Bach tensor is fourth order, trace-free, and obeys the exact linearized Noether identity. Eliminating the constraint on the Einstein subbranch reproduces the Regge–Wheeler equation for $\Psi = Bh_1/r$,

$$B(B\Psi)' + (\omega^2 - V_{\text{RW}})\Psi = 0, \quad V_{\text{RW}} = B\left(\frac{6}{r^2} - \frac{6m}{r^3}\right). \quad (15)$$

Corollary 6.2 (Axial composition). *For the certified axial carrier, parity gives $\delta R = 0$, so*

$$\boxed{\delta B_{ab} = \frac{1}{2}\square(\delta R_{ab}) + C_{acbd}(\delta R)^{cd}.} \quad (16)$$

Consequently the Einstein kernel $\delta R_{ab} = 0$ injects into the Bach kernel, while its curvature image obeys

$$\frac{1}{2}\square\psi_{ab} + C_{acbd}\psi^{cd} = 0, \quad \psi_{ab} := \delta R_{ab}. \quad (17)$$

Let $\mathcal{C}_{\text{ax}}$ be the fixed axial carrier and define

$$\mathcal{H}_B = \{h \in \mathcal{C}_{\text{ax}} : \delta B[h] = 0\}, \quad \mathcal{H}_E = \{h \in \mathcal{H}_B : \delta R[h] = 0\}, \quad (18)$$

$$\mathcal{H}_R = \ker\left(\frac{1}{2}\square + C\circ\right) \cap \text{im}(\delta R|_{\mathcal{C}_{\text{ax}}}). \quad (19)$$

Then the invariant statement supplied by the composition is the exact sequence

$$0 \longrightarrow \mathcal{H}_E \longrightarrow \mathcal{H}_B \xrightarrow{\delta R} \mathcal{H}_R \longrightarrow 0. \quad (20)$$

It does *not* supply a canonical direct sum. A metric lift h_X of $\psi \in \mathcal{H}_R$, when it exists in the chosen domain, may be shifted by any $h_E \in \mathcal{H}_E$. Global lift existence, boundary conditions, and a gauge-independent splitting must therefore be proved separately. In particular, “additional branch” below refers either to the curvature image $\mathcal{H}_R$ or to an explicitly constructed lift, never to an assumed canonical complement.

The identity is componentwise and off shell in the perturbation. It does not extend in this two-term form to the certified non-Einstein static fixture. Thus the clean composition is a Ricci-flat-background theorem.

7 The Ricci carrier reaches the future horizon

In ingoing coordinates (v, r) , Fourier modes are written $e^{i\omega v}$. The extra carrier’s divergence constraint solves its third component polynomially; the apparent Schwarzschild-coordinate pole is a chart artifact.

Theorem 7.1 (Analytic ingoing curvature family). *For axial $\ell = 2$ and $\omega \in \mathbb{R} \setminus \{0\}$, the future horizon $r = 2m$ is a regular singular point of the extra first-order radial system. Its residue spectrum is*

$$\{0, 0, -4im\omega, -2 - 4im\omega\}, \quad (21)$$

and the zero eigenspace has dimension two. Hence there is a two-parameter family of ψ -solutions analytic at the future horizon, with no leading logarithm at the zero exponent.

The restriction excludes $\omega = 0$, where the algebraic multiplicity of the zero residue exponent changes, and makes no claim at resonant complex frequencies. Most importantly, the theorem counts the curvature carrier, not a quotient of metric solutions. The currently certified dimensions are:

Space	Ingoing/analytic dimension	Status
Einstein Regge–Wheeler carrier	1	second-order control
Ricci carrier ψ	2	exact residue calculation
Full Bach metric solution space	not certified	reconstruction/domain open
Metric quotient by the Einstein kernel	not certified	lift and gauge open

Thus future-horizon regularity does not force $\psi = 0$. Explicit metric lifts are constructed in the flux fixtures below, but the exact sequence (20) has not yet been split globally.

8 Action-derived horizon flux

Substituting two on-shell Regge–Wheeler solutions into the current (3) gives

$$F_{EE}^r = -\frac{192\pi\alpha(\omega_1^2 - \omega_2^2)}{5\omega_1\omega_2 r} \Psi_1 \Psi_2. \quad (22)$$

For conjugate pairs $\omega_2 = \pm\omega_1$, this vanishes identically.

Theorem 8.1 (Exact Einstein isotropy). *On Schwarzschild in axial $\ell = 2$, the pure-Weyl Lee–Wald pairing of an Einstein Regge–Wheeler mode with its conjugate vanishes exactly:*

$$F^r(E_\omega, \overline{E_\omega}) = 0. \quad (23)$$

Thus the Einstein line is isotropic in the presymplectic structure inherited from the pure C^2 action. This statement does not identify the Einstein symplectic form with the ordinary Einstein–Hilbert one.

Proposition 8.2 (Controlled horizon-pairing fixtures). *For $m = 1$, axial $\ell = 2$, and conjugate-frequency Hermitian pairings, order-16 ingoing series give*

$$F^r(E_\omega, \overline{X_\omega}) \neq 0, \quad F^r(X_\omega, \overline{X_\omega}) \neq 0 \quad (24)$$

at producer frequencies $\omega = 3/5, 2/7$; an independent verifier repeats the nonvanishing gates at $\omega = 1/2$. At the two stored frequencies the values, divided by $\pi\alpha$ and sampled at two interior radii, are approximately

ω	$F^r(X, \bar{X})/(\pi\alpha)$	$F^r(E, \bar{X})/(\pi\alpha)$
3/5	$-330.58i, -329.98i$	$-218.17 + 43.55i, -215.97 + 51.53i$
2/7	$-71.92i, -71.95i$	$-162.82 + 27.99i, -163.92 + 33.03i$

The exact Regge–Wheeler null identity is the in-run control. These data are controlled numerical fixtures: they prove neither symbolic nonvanishing for all real ω nor an interval-certified error bound.

For a chosen lift basis (E, X) , the horizon block has the form

$$\Omega_h = \begin{pmatrix} 0 & a \\ -\bar{a} & b \end{pmatrix}. \quad (25)$$

Under the admissible lift change $X \mapsto X + cE$, the cross entry a is invariant while b changes. The robust fixture-level conclusions are therefore Einstein isotropy, nonzero cross pairing, and full rank of the tested two-dimensional block. The sign of iF_{XX}^r for one chosen lift is basis dependent and is not promoted to a branch signature or Hilbert norm.

The stored report also records a repaired calculation: an earlier flux evaluation dropped the Fourier factor $e^{i\omega t}$, causing all time derivatives to vanish silently. The exact Regge–Wheeler null control exposed the defect; all displayed fixtures use the repaired full-time current.

9 Asymptotic symbol and endpoint nonselection

At real frequency and large radius, the extra carrier has dispersion

$$(\lambda_r^2 - \omega^2)^2 = 0 \quad (26)$$

and growth exponents

$$\sigma \in \{\pm 2im\omega, \pm 2im\omega - 1\}. \quad (27)$$

The simple-root reading would identify Coulomb logarithmic phases with amplitudes of order r^0 and r^{-1} . But the characteristic is *double*. The polynomial alone does not determine geometric multiplicity. As in $(D^2 + \omega^2)^2 h = 0$, a Jordan chain may produce $r_* e^{\pm i\omega r_*}$ even though every characteristic exponent is oscillatory. The axial Ricci-carrier recurrences are known to all formal orders for every integer $\ell \geq 2$ and real $\omega \neq 0$. The earlier generic- ℓ metric lifts, however, did not impose the independent $v\varphi$ Ricci row. The complete metric endpoint module is certified here only at $\ell = 2$ on the real pilot interval. In the polar calculation below we use only the maximal restriction-stable module through the certified depth. Two additional terminal-only prefix vectors still require their next compatibility test and are not promoted to formal solutions.

Theorem 9.1 (Endpoint regularity-and-leading-symbol nonselection). *On the declared Schwarzschild axial $\ell = 2$, nonzero-real-frequency mode carrier, future-horizon analyticity does not imply $\delta R_{ab} = 0$. Moreover, the presently certified leading characteristic polynomial and formal exponents do not supply an outer condition that distinguishes the Ricci carrier from the Einstein characteristic. Hence these tested local endpoint diagnostics do not select the Einstein kernel.*

This deliberately does *not* say that every ordinary asymptotically flat radiation condition admits an additional metric mode. That stronger statement requires convergence or another completion of the formal series, falloff in a regular tetrad or asymptotic gauge, a differentiable Lee–Wald phase space, and a horizon-to-infinity connection map. None is supplied by the formal calculations below.

9.1 Complete axial reconstruction on the $\ell = 2$ pilot

The earlier generic-angular metric audit imposed the $x\varphi$ and $r\varphi$ reconstruction equations but omitted the independent $v\varphi$ equation. Put

$$C = \frac{\delta R_{v\varphi}}{S_2} - P. \quad (28)$$

On every solution of the two differential rows and the Ricci-carrier equations, exact differentiation gives

$$C' = -\frac{2}{r}C, \quad \kappa = r^2 C = \text{constant}. \quad (29)$$

The complete reconstruction is therefore the $\kappa = 0$ fibre. Solving this algebraic constraint for H_0 produces an exact six-state flow on $(P, P', Q, Q', H_1, H'_1)$ and the short exact sequence

$$0 \longrightarrow \mathcal{E}_{\text{ker}}^2 \longrightarrow \mathcal{E}_{\text{Bach,ax}}^6 \longrightarrow \mathcal{E}_{\text{Ricci}}^4 \longrightarrow 0. \quad (30)$$

It supplies six independent formal columns at each endpoint: four carrier lifts and two complete Einstein-kernel columns. This statement is exact for $\ell = 2$, $M = 1$, and real $\omega \in [1/2, 3/4]$; it is not a generic- ℓ metric theorem.

The legacy Phase-2 X_0 has

$$C_{X_0} = \frac{i(\omega - 18i)}{2\omega^2} r^{-2}. \quad (31)$$

It becomes a complete non-Einstein lift after an exact transverse addition. The vector formerly labelled E_0 is one half of that transverse vector and is not itself a complete Einstein solution; all claims using it as an Einstein representative are superseded.

For the repaired X_0 , the literal current audit has no coefficient at any nonintegrable power $p \geq -1$. Its first self coefficient is

$$F^v(X_0, \overline{X}_0) = \frac{32i\pi\alpha_W(540 - \omega^2)}{15\omega^3(\omega^2 + 4)} r^{-2} + O(r^{-3}), \quad (32)$$

which is nonzero on the pilot interval. The same finite class is preserved under the true rate-zero Einstein-kernel shear. It is *not* preserved under an arbitrary Einstein shear: the true oscillatory Einstein kernel has cross term

$$F^v(E_{\text{osc}}, \overline{X}_0) = \frac{48\pi\alpha_W\omega^3(4\omega + i)}{5} e^{-2i\omega r} r^{3-4i\omega} + \dots, \quad (33)$$

with a nonzero coefficient throughout the real pilot interval.

Theorem 9.2 (Scoped axial reconstruction and radial disposition). *For $\ell = 2$, $M = 1$, and real $\omega \in [1/2, 3/4]$, the complete axial formal endpoint module has dimension six. Its repaired X_0 direction is non-Einstein and radially finite in the fixed representative and under true rate-zero Einstein shears, while true oscillatory Einstein shears produce divergent cross terms. Thus the pilot supplies a finite formal non-Einstein direction but no representative-independent axial selection or quotient theorem.*

This is an exact formal asymptotic coefficient audit. It establishes no convergence or endpoint wave-packet flux space *by itself*, and it establishes no horizon-to-infinity matching, global scattering channel, stability, or CPT property. The endpoint flux construction below is a separate completion of one part of that boundary problem. The other axial additional direction and the generic angular tower require separate all-row audits.

9.2 Exact axial null-endpoint wave-packet flux

The complete axial pilot admits exact matching trace spaces

$$\mathcal{X}_{\mathcal{J}^-} = \text{span}\{\text{XI0}, \text{XI1}, \text{EI0}\}, \quad \mathcal{X}_{\mathcal{J}^+} = \text{span}\{\text{XI2}, \text{XI3}, \text{EI2}\}. \quad (34)$$

Positive-frequency amplitudes lie initially in $C_c^\infty([1/2, 3/4]; \mathbb{C}^3)$, with negative frequencies fixed by the real-field involution; the completion is $L^2([1/2, 3/4]; \mathbb{C}^3)$ at each endpoint. Pulling the frozen action-derived current from Schwarzschild t coordinates to the ingoing Eddington–Finkelstein

reconstruction, including differentiated reconstruction terms, gives Stokes-oriented Hermitian Grams G_- and G_+ . After division by $\pi\alpha_W$,

$$\det G_- = \frac{14155776}{125}\omega^3, \quad \det G_+ = \frac{3538944}{125}\omega. \quad (35)$$

The past sign includes the reversal between the coordinate-radial direction and the outward Stokes orientation at $\mathcal{J}^-$.

Theorem 9.3 (Scoped axial null-endpoint flux). *For strict pure Weyl gravity with $M = 1$, axial $\ell = 2$, and $\omega \in [1/2, 3/4]$, both exact wave-packet endpoint trace spaces have dimension three. Their action-derived Stokes Grams have rank three, zero radical, no frequency wall, and*

$$\text{inertia}(G_-) = \text{inertia}(G_+) = (1, 2, 0) \quad (\alpha_W > 0).$$

Reversing the overall action sign reverses the positive and negative counts.

The signature has a simple exact anatomy. With the displayed Stokes orientations and after division by $\pi\alpha_W$, at $\mathcal{J}^-$ set

$$E_- = \text{EI}0, \quad X_- = \text{XI}0, \quad Y_- = -\omega^2 \text{XI}0 - 2i\omega \text{XI}1 + \text{EI}0.$$

Then

$$G_-(E_-, E_-) = 0, \quad G_-(E_-, X_-) = \frac{384\omega}{5}, \quad G_-(Y_-, Y_-) = -\frac{384\omega^3}{5},$$

and Y_- is orthogonal to E_- and X_- . Likewise, at $\mathcal{J}^+$ set

$$E_+ = \text{EI}2, \quad X_+ = \text{XI}3, \quad Y_+ = i\omega \text{XI}2 + 4(4\omega^2 - i\omega - 2)\text{XI}3.$$

Here

$$G_+(E_+, E_+) = 0, \quad G_+(E_+, X_+) = \frac{96\omega}{5}, \quad G_+(Y_+, Y_+) = -\frac{384\omega}{5},$$

with the same orthogonality. The Einstein–additional planes have determinants $-147456\omega^2/25$ and $-9216\omega^2/25$, respectively. Thus each endpoint has the exact Witt decomposition

$$(1, 2, 0) = (1, 1, 0) \oplus (0, 1, 0).$$

This identifies a hyperbolic Einstein–additional plane and a separate negative line. It does not identify the additional column as a spectral derivative or time-translation Jordan partner. Indeed, the complete six-dimensional axial module is not conjugate by an endpoint-regular invertible change of variables to a scalar Regge–Wheeler square: the latter has only four local columns, whereas the certified exact sequence has dimensions $2 + 4 = 6$. Exact cyclic elimination of the four-dimensional Ricci carrier instead gives

$$L_4 = L_x \circ L_{\text{RW}},$$

and exact embedding and quotient maps certify

$$0 \longrightarrow M_{\text{RW}} \longrightarrow M_{A_4} \longrightarrow M_x \longrightarrow 0.$$

The carrier-zero metric kernel is independently equivalent to M_{RW} . After the rational rescaling $Z = y/[r^2(r-2)]$, the quotient equation is

$$D_{r*}^2 y + 2i\omega D_{r*} y - \frac{6(r-2)}{r^3} y = 0,$$

the $\ell = 2$ spin-one Regge–Wheeler equation in the ingoing-EF convention. Thus the full module has a certified triangular filtration with diagonal factors $M_{\text{RW}}, M_{\text{RW}}, M_x$. The natural exact gauge retains a nonzero extension term, so a direct $\ker L_{\text{RW}}^2 \oplus \ker L_x$ splitting and a time-translation Jordan interpretation are not claimed.

The incoming quotient map nevertheless resolves the Witt signs by differential factor. In the basis (XI0, XI1, EI0), it obeys

$$\pi_x(\text{XI0}) = 2, \quad \pi_x(\text{XI1}) = -2i\omega, \quad \pi_x(\text{EI0}) = 0.$$

Set

$$R = \text{XI0} - \frac{i}{\omega}\text{XI1}, \quad S = -\omega^2\text{XI0} - 2i\omega\text{XI1} + \frac{4}{3}\text{EI0}, \quad E = \text{EI0}.$$

Then $\ker \pi_x = \text{span}\{E, R\}$, while $\pi_x(S) = -6\omega^2$. In the ordered basis (R, S, E) , the exact incoming Gram divided by $\pi\alpha_{\text{W}}$ is

$$\begin{pmatrix} 624/(5\omega) & 0 & 576\omega/5 \\ 0 & -384\omega^3/5 & 0 \\ 576\omega/5 & 0 & 0 \end{pmatrix}.$$

Consequently the metric/carrier spin-two extension has inertia $(1, 1, 0)$, and its orthogonal spin-one quotient lift is strictly negative, of inertia $(0, 1, 0)$. Equivalently, after $R \mapsto R - 13E/(24\omega^2)$, this is the canonical orthogonal sum

$$\begin{pmatrix} 0 & 576\omega/5 \\ 576\omega/5 & 0 \end{pmatrix} \hat{\oplus} (-384\omega^3/5).$$

This factor attribution does not require the differential extension to split. With the quotient normalized to one,

$$S_{\text{unit}} = -\frac{S}{6\omega^2}, \quad \pi_x(S_{\text{unit}}) = 1, \quad \frac{G_-(S_{\text{unit}}, S_{\text{unit}})}{\pi\alpha_{\text{W}}} = -\frac{32}{15\omega}.$$

In this canonical basis the fiberwise form is congruent to $\text{diag}(1, -1, -1)$. Across the whole positive frequency axis its natural Hilbert majorant is weighted:

$$\int_0^\infty \left[\frac{576\omega}{5}(|e|^2 + |r|^2) + \frac{32}{15\omega}|s|^2 \right] d\omega.$$

Thus a finite-flux spin-one amplitude must obey $\int_0^\epsilon |s(\omega)|^2 d\omega/\omega < \infty$ near threshold. On the positive-frequency Fourier core these are the homogeneous fractional Sobolev weights

$$(e, r) \in \dot{H}^{1/2}, \quad s \in \dot{H}^{-1/2},$$

up to Fourier normalization and the real-field extension. Weighted integrability alone does not define $s(0)$; if a continuous threshold trace is separately available, it must vanish. The unweighted L^2 topology remains equivalent on compact frequency bands away from zero, while the static/memory sector at $\omega = 0$ is separate. Pointwise invertibility of T_- does not by itself give a bounded global operator on this weighted direct integral. In weight-normalized factor frames, write

$$T_- = \begin{pmatrix} a_2 & b & c \\ 0 & a_2 & d \\ 0 & 0 & a_1 \end{pmatrix}, \quad a_s = A_{\text{in},s}.$$

Since the scalar Wronskians give $|a_s^{-1}| \leq 1$ on the positive real axis, boundedness of the incoming-to-horizon population map T_-^{-1} reduces to weighted bounds on only

$$\frac{b}{a_2^2}, \quad \frac{d}{a_2 a_1}, \quad \frac{bd - a_2 c}{a_2^2 a_1}.$$

Threshold, high-frequency, and limiting-absorption estimates for these three extension ratios are the remaining quantitative gate between separated-mode no-growth and time-domain boundedness.

The normalized signature basis also defines the canonical classical fundamental symmetry $C_{\text{fac}} = \text{diag}(1, -1, -1)$. In the null basis it exchanges EI and RI_0 , negates SI_{unit} , and yields the positive majorant $\text{diag}(a, a, b)$. This is an endpoint Krein-space fundamental symmetry, not a Mannheim C operator: BRST descent and compatibility with the full scattering dynamics are not established.

The L^2 completion here uses the auxiliary positive coefficient norm, not the indefinite flux as a norm. Exact Sturm counts applied to $\|G_{\pm}(\omega)\|_F^2$ and $\|G_{\pm}(\omega)^{-1}\|_F^2$ give the uniform normalized estimate

$$\|a\|_{L^2} \leq \|G_{\pm} a\|_{L^2} \leq 645 \|a\|_{L^2}. \quad (36)$$

Thus the endpoint forms are bounded and uniformly nondegenerate, rather than merely pointwise invertible. The continuous fundamental symmetry $J_{\pm} = \text{sign}(G_{\pm})$ and positive majorant $|G_{\pm}|$ make each endpoint a Krein space with topology equivalent to the auxiliary L^2 topology. This majorant is not a CPT metric, positive energy, or particle norm. The exact matrices and separate endpoint bounds are printed in Appendix A.

The past-boundary sign is already absorbed into the incoming Gram. Hence a future global relation $a_{\text{out}} = T a_{\text{in}}$ must obey

$$T^{\dagger} J_{\text{out}} T = J_{\text{in}}; \quad (37)$$

without horizon channels this would reduce to $T^{\dagger} G_+ T = G_-$. No complete outgoing-plus-horizon scattering map is constructed here. The theorem below does, however, prove that the full incoming $\mathcal{J}^-$ form is globally populated by future-horizon-regular solutions.

For the presymplectic ambiguity $\theta \mapsto \theta + \delta Y + dZ$, the δY term vanishes by $\delta^2 = 0$, and angular exact terms integrate to zero. A stationary, globally defined local finite-tangential-jet dZ with finite trace-only pullback contributes only cut terms; these vanish on the C_c^{∞} frequency core because the inverse Fourier traces are Schwartz, and hence on its continuous L^2 extension. This scoped statement does not cover radial or subleading, nonlocal or soft, explicit-time, or nondecaying improvements. Their additive $\Delta G_{\pm}$ and inertia strata remain open.

Proposition 9.4 (Index pullback and local wave-packet consequence). *Fix either certified axial endpoint and a frequency at which its Hermitian flux form $G(\omega)$ has inertia $(1, 2, 0)$. Let*

$$T_{\omega} : U_{\omega} \longrightarrow \mathcal{X}_{\omega}$$

be any globally matched, gauge-reduced trace map, and set $K_{\omega} = T_{\omega}^{\dagger} G(\omega) T_{\omega}$. After quotienting $\ker T_{\omega}$, write $r = \text{rank } T_{\omega}$. Then

$$n_-(K_{\omega}) \geq r - 1.$$

In particular, a populated endpoint image of rank at least two necessarily contains a negative-flux direction, while full rank forces inertia $(1, 2, 0)$.

If T_{ω} and the horizon-regular frame vary continuously near an interior frequency ω_0 , then $\text{rank } T_{\omega_0} \geq 2$ implies the existence of a nonzero smooth compact-frequency horizon amplitude whose populated endpoint flux is strictly negative. In fact the negative index of the populated wave-packet space is then infinite.

Proof. Let $W = \text{im } T_\omega$. If the restriction of G to W has inertia (a, b, c) , it has a nonnegative subspace of dimension $a + c$. Projection of such a subspace onto the one-dimensional positive spectral subspace of G is injective: a vector with zero positive projection has nonpositive norm in the negative spectral subspace, and nonnegativity forces it to vanish. Hence $a + c \leq 1$, and $b = r - a - c \geq r - 1$.

For the second statement choose $u \in U_{\omega_0}$ with $u^\dagger K_{\omega_0} u < 0$. In a continuous local trivialization the same coefficient vector has negative flux on some open interval $J \ni \omega_0$. Multiplying it by any nonzero $\chi \in C_c^\infty(J)$ and integrating the diagonal frequency-space pairing gives a strictly negative finite wave-packet flux. Choose countably many nonzero cutoffs with pairwise disjoint supports inside J . Their cross-pairings vanish, while every self-pairing is negative. They therefore span an infinite-dimensional negative subspace. $\square$

The first global gate can in fact be closed analytically for the incoming endpoint. Let

$$T_-(\omega) : \text{span}\{\text{XH0a}, \text{XH0b}, \text{EH0}\} \longrightarrow \text{span}\{\text{XI0}, \text{XI1}, \text{EI0}\}$$

map future-horizon-regular coefficients to incoming $\mathcal{I}^-$ coefficients.

Theorem 9.5 (Global population of the axial incoming trace). *For every real $\omega > 0$,*

$$T_-(\omega) \in GL(3, \mathbb{C}).$$

In factor-adapted endpoint frames its determinant is

$$\det T_-(\omega) = -\frac{(2\omega - i)(4\omega - i)^2}{4(\omega - i)} A_{\text{in},2}(\omega)^2 A_{\text{in},1}(\omega),$$

where $A_{\text{in},2}$ and $A_{\text{in},1}$ are the incoming Jost coefficients for the spin-two and spin-one real short-range potentials

$$V_2 = \frac{6(r-2)(r-1)}{r^4}, \quad V_1 = \frac{6(r-2)}{r^3}.$$

Proof. The exact module filtration and endpoint quotient maps give one horizon-ingoing and one incoming-infinity Jost line for each of the three diagonal factors. The changes from the repository endpoint frames to these factor-adapted frames are triangular with determinant one. The resulting connection is triangular and its diagonal entries are the two spin-two and one spin-one Jost connection coefficients, multiplied by the displayed nonzero endpoint normalizations. For a unit horizon-ingoing solution of either real potential, Wronskian conservation gives

$$|A_{\text{in},s}|^2 - |A_{\text{out},s}|^2 = 1, \quad s = 1, 2.$$

Hence $A_{\text{in},s} \neq 0$. The rational prefactor is also nonzero, since its squared modulus is

$$\frac{(4\omega^2 + 1)(16\omega^2 + 1)^2}{16(\omega^2 + 1)}.$$

The cyclic and triangular-map determinants are nonzero for $r > 2$ and real $\omega > 0$; the horizon recurrence pivots do not vanish, their integer-spaced resonance is compatible, and the two infinity rates remain distinct. Thus no endpoint-map or Frobenius collision interrupts the argument on the positive real axis. The zero-frequency sector is separate. $\square$

Moreover, on the original pilot interval the determinant has the uniform exact margin

$$|\det T_-(\omega)| \geq \sqrt{\frac{5}{2}}.$$

Indeed, after $x = \omega^2$, the derivative of the squared rational prefactor is

$$\frac{(16x+1)(128x^2+208x+35)}{16(x+1)^2} > 0,$$

and $|A_{\text{in},s}| \geq 1$. This is a volume-determinant bound, not by itself a smallest-singular-value estimate. On the full positive real axis the same monotonicity gives

$$|\det T_-(\omega)| > \frac{1}{4}, \quad \omega > 0,$$

with $1/4$ approached only as $\omega \rightarrow 0^+$.

Congruence by the invertible T_- preserves inertia. The horizon-regular pullback of the incoming endpoint form therefore has inertia $(1, 2, 0)$ at every real positive frequency, and its wave-packet completion on every compact positive-frequency interval has infinite positive and negative index. In particular, the conditional conclusion of Proposition 9.4 is activated for $\mathcal{J}^-$: the incoming indefinite directions are genuine globally populated radial channels and cannot be removed by finitely many global charge conditions, a finite-dimensional residual quotient, or deletion of finitely many exceptional frequencies.

This is a one-sided global connection and flux theorem, not yet a complete scattering theorem. It does not determine the rank of the outgoing $\mathcal{J}^+$ reflection block T_+ , the reflection amplitudes, non-real complex-frequency poles, total canonical energy, a particle norm, CPT positivity, time-domain stability, or invariance under unrestricted radial or corner improvements of the presymplectic potential. It does exclude a future-horizon-regular solution with zero incoming $\mathcal{J}^-$ trace, and hence excludes purely outgoing real-frequency axial modes and real-axis incoming spectral singularities for every $\omega > 0$.

There is also a scoped complex-frequency result. The repository phase is $e^{+i\omega t}$, so exponential growth corresponds to $\text{Im } \omega < 0$. A zero of either reduced incoming Jost coefficient in that half-plane would produce an $L^2(dr_*)$ mode of

$$-D_{r_*}^2 + V_s, \quad s = 1, 2,$$

with eigenvalue ω^2 . Both displayed potentials are nonnegative. Self-adjointness excludes a nonreal eigenvalue when $\text{Re } \omega \neq 0$, and nonnegativity excludes $\omega^2 < 0$ on the imaginary axis. The exact inclusion, quotient, and master maps preserve the future-horizon-regular and pure-outgoing germ classes throughout that half-plane. Successive quotient elimination therefore proves that the complete non-split six-state axial $\ell = 2$ system has no growing separated mode. Equivalently, its intrinsic factor Evans product

$$\mathcal{E}_{\text{reg}}(\omega) = A_{\text{in},2}(\omega)^2 A_{\text{in},1}(\omega)$$

is zero-free for $\text{Im } \omega < 0$. This is a separated-mode spectral no-growth theorem, not time-domain boundedness, decay, completeness, or a full PDE stability theorem. The frame events at $\omega = i/4, i/2, i$ lie in the temporally damped half-plane in this convention and remain unclassified by the energy argument.

The non-split spin-two extension can also be extracted exactly in local Regge–Wheeler companion frames. Its rational 2×2 source matrix $\mathcal{E}_{\text{RW}}$ has rank one, but it is not a pointwise scalar multiple of $\partial_\omega A_{\text{RW}}$. More strongly, a putative global rational-gauge identity

$$\mathcal{E}_{\text{RW}} = q(\omega)\partial_\omega A_{\text{RW}} + B' + BA_{\text{RW}} - A_{\text{RW}}B$$

has trace residue $4iq$ at $r = 2$, whereas a rational derivative has zero residue. It therefore forces $q = 0$ in that gauge class. This does not exclude endpoint-analytic logarithmic spectral-phase gauges.

At a simple damped spin-two zero, the invariant scattering datum remains $[c] \in \mathcal{O}/(A_{\text{in},2})$. The congruence $c \equiv qA'_{\text{in},2} \pmod{A_{\text{in},2}}$ with unspecified q is not predictive: $[A'_{\text{in},2}]$ is a unit, so such a class $[q]$ always exists. Selecting the local Smith type requires a certified QNM and adjoint germ together with a convergent Fredholm pairing; no nonzero extension resonance is promoted here.

Proposition 9.6 (Local QNM Smith dichotomy). *Let ω_n be a simple zero of the spin-two incoming Jost coefficient $a = A_{\text{in},2}$, and suppose the spin-one coefficient $f = A_{\text{in},1}$ is nonzero there. After analytic elimination of the spin-one factor, the local connection is equivalent to*

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}.$$

If $[b] \neq 0$ in $\mathcal{O}_{\omega_n}/(a)$, its Smith valuations are $(0, 2)$: the resonance has one root vector, a length-two chain, and a second-order resolvent pole. If $[b] = 0$, the valuations are $(1, 1)$: there are two independent root vectors and only simple poles. Under compatible endpoint normalizations the selector is the invariant Fredholm overlap

$$\beta_n = \langle \tilde{u}_n, \mathcal{E}_{\text{RW}}(\omega_n)u_n \rangle, \quad [b] \neq 0 \iff \beta_n \neq 0.$$

The present certificate proves this dichotomy and commutator invariance, but does not evaluate β_n at any physical QNM.

Theorem 9.7 (Exact future-horizon flux). *For axial $\ell = 2$ and every real $\omega > 0$, the exact outward-oriented future-horizon Gram $H_{\mathcal{H}+}$, evaluated from the correlated Frobenius series and the action current, has rank three and inertia*

$$\text{inertia } H_{\mathcal{H}+} = (1, 2, 0) \quad (\alpha_{\text{W}} > 0).$$

Its metric/carrier spin-two kernel plane has inertia $(1, 1, 0)$, while the orthogonal unit spin-one quotient has exact norm

$$-\pi\alpha_{\text{W}} \frac{32}{15\omega}.$$

Neither $H_{\mathcal{H}+}$ nor $-H_{\mathcal{H}+}$ is positive semidefinite.

The all-positive-frequency extension follows from the symbolic recurrence: its residue eigenbasis and all factor-frame denominators are collision-free on $\omega > 0$, the three integer resonances have exact zero residual, and the omitted-head power count cannot change the Laurent constant.

The oriented Stokes identity is therefore

$$H_{\mathcal{H}+} + T_+^\dagger G_+ T_+ - T_-^\dagger G_- T_- = 0.$$

The horizon theorem independently reproduces the hyperbolic-spin-two plus negative-spin-one anatomy of the incoming endpoint. Its indefiniteness also means that a semidefinite shortcut cannot determine rank T_+ ; the outgoing connection remains necessary.

There is nevertheless an exact conditional Krein theorem. Normalize the incoming, outgoing, and horizon forms either to

$$J_\sigma = \text{diag}(1, -1, -1) \quad \text{or} \quad J_0 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The signature frame J_σ carries the invariant sign decomposition; the null frame J_0 carries factor triangularity. They must not be conflated. If a typed, same-frame T_+ and the global Stokes defect are supplied, then with normalized maps

$$R = \widehat{T}_+ \widehat{T}_-^{-1}, \quad A = \widehat{T}_-^{-1}, \quad S = \begin{pmatrix} R \\ A \end{pmatrix},$$

one has

$$S^\dagger(J \oplus J)S = J, \quad D := J - R^\dagger J R = A^\dagger J A.$$

Thus D is nondegenerate with inertia $(1, 2, 0)$, and S admits a noncanonical algebraic completion to $U(2, 4)$. The determinant is normalization-sensitive:

$$\det D = |\det \widehat{T}_-|^{-2} = |A_{\text{in},2}|^{-4} |A_{\text{in},1}|^{-2}.$$

Using the raw T_- without the endpoint normalizer ratio is incorrect. Equivalently, with the Krein adjoint $M^\sharp = J^{-1} M^\dagger J$, the conditional identity is

$$R^\sharp R + A^\sharp A = I.$$

Because A is invertible, $I - R^\sharp R$ is invertible and hence $1 \notin \sigma(R^\sharp R)$. This does not exclude nonzero horizon waves of zero scalar flux: a nondegenerate indefinite horizon form still has a null cone.

This conditional theorem is not yet a physical scattering result. The repository does not certify the typed full 3×3 T_+ , its same-frame normalization, or the orientation-correct global Stokes defect. It likewise does not certify a three-parameter (α, γ, μ) reduction of the null-frame extension entries.

For comparison, the Einstein problem admits a global finite-energy Schwarzschild scattering theory relating initial data and radiation spaces by Hilbert-space isomorphisms [23]. Flat-space conformal-higher-spin scattering also admits conformal-gravity states beyond ordinary Einstein gravitons [24]; it does not construct the Schwarzschild Bach connection considered here.

9.3 Polar mixed finite line and its exceptional wall

On the maximal restriction-stable polar module, choose the ordered formal basis (E, X_0, X_1, X_2) . The denominator-cleared current matrices at powers $p = 0, -1, -2$ are anti-Hermitian. The $p = 0$ matrix has rank three and a one-dimensional mixed Einstein/additional radical. If y spans that line, then

$$y^\dagger N^{(-1)} y = 0, \tag{38}$$

while its first finite coefficient factors as

$$y^\dagger N^{(-2)} y = C \omega^{51} \Lambda^3 (\Lambda - 2)^5 P_6^2 P_{20} Q_{21} (\Lambda, \omega^2). \tag{39}$$

The constant C is nonzero, and the exact sum-of-squares and resultant certificates show that $P_6 P_{20} \neq 0$ for integer $\ell \geq 2$ and real $\omega \neq 0$. Therefore the filtered line is generically nonradical. At $Q_{21} = 0$ its deeper disposition remains open.

Writing $x = \widehat{\omega}^2 > 0$, the exact positive-root phase diagram is

Harmonic	positive x -roots	real nonzero $\widehat{\omega}$ -roots
$\ell = 2$	0	0
$\ell = 3$	3	6
$4 \leq \ell \leq 10$	1	2
$11 \leq \ell \leq 40$	3	6
$\ell \geq 41$	1	2

At the legacy fixture $(\Lambda, x) = (6, 9/25)$,

$$Q_{21}(6, 9/25) = -\frac{174226120816040380076641138108451235935620694016}{227373675443232059478759765625} \neq 0. \quad (40)$$

The previously recorded longer rational was $Q_{21}(6, 81/625)$, caused by squaring x twice; it was neither the fixture value nor a Hilbert norm.

Theorem 9.8 (Generic polar formal radial disposition). *Away from the explicit Q_{21} wall, a one-dimensional mixed Einstein/additional polar line is finite through the nonintegrable layers and has nonzero first finite pairing. Thus formal radial pairing finiteness does not canonically select a polar Einstein representative.*

This theorem neither extends the terminal-only polar prefixes nor matches the mixed line to future-horizon-regular data. In particular, the axial and polar infinity results do not identify a global solution admitted at both ends of the exterior.

9.4 Local Cauchy selection remains available

There is no no-go against local causal selection. On the globally hyperbolic exterior with Cauchy surface $\Sigma = \{t = \text{const}, r > 2m\}$, the local Cauchy condition

$$\psi|_{\Sigma} = 0, \quad \nabla_n \psi|_{\Sigma} = 0 \quad (41)$$

propagates $\psi = 0$ in the axial sector unconditionally: the carrier operator equals $\frac{1}{2}\square + C \circ (-)$ exactly there (the trace vanishes identically on the class), is normally hyperbolic, and its Bianchi constraint obeys the exact transport identity $\nabla^a (L\psi)_{a\varphi} = \frac{1}{2}\square B_{\varphi}$. In the polar sector the operator is identically traceless at the PDE level, the conformal direction is uncontrolled, and $\psi_{\text{conf}}(t^4 \chi(r) P_2)$ is an exact nonuniqueness witness with vanishing Cauchy data; the certified conformal trace relation $S(\psi_{\text{conf}}(\Phi)) = -3\square\Phi$ then reduces any zero-data solution to the conformal-gauge orbit, so local Cauchy data selects the Einstein image exactly *modulo conformal gauge* (and exactly on the traceless slice). Dropping the normal-derivative datum is certified to fail by a time-odd mode witness. The open physical questions are whether this restriction is preserved by nonlinear interactions and whether it follows from a differentiable asymptotic phase space rather than being imposed by hand.

10 Polar chain: composition, reach, current, and disposition

For polar $\ell = 2$ Regge–Wheeler-gauge perturbations, the trace variation δR need not vanish. The axial formula acquires two universal trace terms.

Corollary 10.1 (Polar trace-coupled composition). *On the certified polar $\ell = 2$ carrier, Proposition 6.1 gives*

$$\delta B_{ab} = \frac{1}{2}\square(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a \nabla_b \delta R - \frac{1}{12}g_{ab}\square\delta R. \quad (42)$$

The Einstein polar kernel injects by $\delta R_{ab} = 0$, while the curvature image obeys a second-order trace-coupled Lichnerowicz system for $(\delta R_{ab}, \delta R)$.

The polar chain is now certified through the same stations as the axial chain. Four structural facts organise it.

(i) *Polar Einstein kernel.* In the t -chart Regge–Wheeler polar gauge the traceless angular row of δR_{ab} equals $(H_0 - H_2)/2$, forcing $H_2 = H_0$; the remaining rows reduce exactly to a two-dimensional first-order system $dY/dr = M(r)Y$, $Y = (K, H_1)$, with H_0 algebraic. In horizon-adapted variables (K, BH_1) the ingoing residue spectrum is $\{0, -4im\omega\}$, matching the axial Regge–Wheeler control. A Schrödinger-form master scalar with ω -free potential was *not* found within the searched bounded rational ansatz classes; that anchor is recorded fail-closed open and nothing below depends on it.

(ii) *Conformal gauge and polar reach.* The polar carrier system on the Bianchi-constrained $(\delta R_{ab}, \delta R)$ is exactly three second-order equations in four radial functions: the operator (42) is identically traceless and divergence-free on the constraint surface, and the one-function underdeterminacy is linearized conformal gauge ($h = \Phi g$ is pure gauge and $\psi_{\text{conf}} = \delta \text{Ric}[\Phi g]$ annihilates every row for arbitrary Φ); this direction is absent in the axial sector. On the algebraic traceless slice the principal part diagonalises and the future horizon is a regular singular point with residue spectrum $\{0, 0, 0, 1 - 4im\omega, -1 - 4im\omega, -3 - 4im\omega\}$. Quotienting the regular conformal-gauge direction leaves a *two*-parameter physical ingoing-analytic polar carrier family at every $\omega \in \mathbb{R} \setminus \{0\}$ — the polar twin of the axial reach theorem.

(iii) *Polar current.* The sphere-integrated action-derived polar bilinear obeys the same off-shell 4α identity. On shell of the two-dimensional polar Einstein system every bilinear coefficient carries the factor $(\omega_1 + \omega_2)$: the polar Einstein self-block is exactly null for conjugate pairs, as in the axial sector. In addition the conformal direction Φg pairs to zero with an *arbitrary* off-shell polar perturbation: the sphere-integrated polar presymplectic form needs no conformal quotient at the bilinear level. Composition fixtures ($\omega = 3/5$, two interior radii, exact arithmetic) certify that the realized Ricci image closes on the full three-dimensional analytic carrier space — every ingoing-analytic polar carrier mode lifts to a metric perturbation with all seven δRic rows verified — and that the Einstein–carrier cross pairing is nonzero. Cross-pairing nonvanishing is representative-independent because the Einstein block is null; the positive carrier self-norms are certified at the canonical composed representatives only, and the representative-invariant sign theory is open.

(iv) *Polar disposition.* The polar carrier’s asymptotic symbol is $(\lambda^2 - \omega^2)^3$: it propagates on the Einstein characteristics, with Coulomb log-phases and amplitude falloffs r^{-1}, r^{-2}, r^{-3} — all decaying at real frequencies, strictly stronger than the axial r^0, r^{-1} case. The tested endpoint diagnostics (future-horizon analyticity, leading outer symbol and falloff class) therefore fail to separate the polar Ricci carrier from the polar Einstein carrier, exactly as in the axial sector: the endpoint-nonselection result is now *parity complete* at the $\ell = 2$ linear mode level. As before, this is not a classification of all local differential boundary operators.

(v) *Polar formal infinity module.* A later full-seven-row audit found that the shallow four-state master used in the earlier norm-selection table contained a source-zero direction that did not satisfy the complete Ricci system. The restriction-stable replacement and its current filtration are summarized by Equation (39): a mixed Einstein/additional line is finite and generically nonradical. The former power-enhanced/logarithmic fixture and parity-complete Einstein-only norm-selection theorem are therefore superseded. The independently verified polar composition, horizon reach, Einstein isotropy, conformal degeneracy, and controlled horizon pairings in items (i)–(iv) are unaffected.

11 Horizon monodromy temperature

One statement of Hawking type is available at the REDUCED-MODE level without any Lorentzian quantum construction. The surface gravity of the certified background is $\kappa = 1/(4m)$ with $T_H = \kappa/2\pi = 1/(8\pi m)$; the geometric-clock value agrees exactly with the certified normalized-frame first-law temperature.

Theorem 11.1 (Monodromy universality). *Under the horizon continuation $\rho \rightarrow e^{2\pi i}\rho$, every certified ingoing residue exponent of the four $\ell = 2$ mode families (axial and polar, Einstein and Ricci carrier) has continuation factor exactly 1 or $e^{8\pi m\omega} = e^{\omega/T_H}$, and every family contains exponents with the nontrivial factor.*

The integer parts of the exponents are monodromy-trivial; the universal $-4im\omega$ part carries the thermal factor. Hence the Boltzmann ratio $|\beta/\alpha|^2 = e^{-\omega/T_H}$ of the Damour–Ruffini continuation is identical for the Einstein and additional curvature families in both parity sectors: at the mode level, the additional carrier is thermally weighted at exactly the Hawking temperature, and by the certified nonzero mixed pairings it participates in the horizon flux with that weight. This is a mode-level (REDUCED-MODE) statement only: no Hadamard state, renormalized stress tensor, grey-body factor, luminosity, back-reaction, or LORENTZIAN-CAUSAL Hawking theorem is claimed, and none exists in the repository.

12 Result ledger

Question	Certified answer	Scope
Static Bach-flat black holes?	exact Laurent-family classification and three-horizon fixture; Einstein sheet corrected	local/static
Differentiable energy?	residual-basic normalized generator, exact static first law	$u \neq 0$; signed orientation; $f(J)$ freedom
Einstein axial waves present?	yes; exact Regge–Wheeler injection	Schwarzschild, axial $\ell = 2$
Additional axial content?	nonzero second-order Ricci carrier; no canonical metric split	same
Reach future horizon?	two analytic ingoing ψ amplitudes	nonzero-real-frequency carrier
Carry current?	Einstein self-block exactly null; mixed/additional blocks nonzero in fixtures	exact identity plus three frequencies
Selected by tested endpoints?	no: horizon and leading-symbol tests do not force $\psi = 0$	not a general causal no-go
Formal radial infinity?	On the complete axial $\ell = 2$ pilot, repaired X_0 is finite in the fixed representative and under rate-zero Einstein shears, but oscillatory Einstein shears cross it divergently; the polar sector has a generic-angular mixed finite line	no representative-invariant axial quotient, phase space, norm, or global matching
Null-endpoint wave-packet flux?	Exact three-dimensional axial L^2 trace spaces at both null ends; Grams have rank three, zero radical, and inertia $(1, 2, 0)$ for $\alpha_W > 0$	$\ell = 2, M = 1, \omega \in [1/2, 3/4]$
Incoming global channels?	Exact RW/RW/spin-one triangular filtration and $T_-(\omega) \in GL(3, \mathbb{C})$ for every real $\omega > 0$; the full incoming indefinite trace is populated	outgoing T_+ , non-real poles, full stability, CPT and complete scattering open
Polar exceptional wall?	generically nonradical finite line; exact wall $Q_{21}(\ell(\ell + 1), \hat{\omega}^2) = 0$ with certified harmonic root counts	deeper filtration open on the wall
Einstein–extra horizon pairing invariant?	Einstein block isotropic; nonzero cross pairing and full rank in controlled fixtures; extra self-pairing sign not invariant	axial $\ell = 2$, real $\omega \neq 0$
Complex-frequency structure?	the complete non-split axial $\ell = 2$ system has no lower-half-plane growing separated mode in the $e^{+i\omega t}$ convention; the selected frames have damped-half-plane events at $\{i/4, i/2, i\}$	time-domain PDE stability, regularized damped poles, and the quasinormal spectrum remain open
Static ($\omega = 0$) sectors?	log-classified: two-parameter log-free families both parities; polar Einstein variables degenerate	classification only; static composition open
Polar composition?	exact trace-coupled identity; realized image closes on the full analytic carrier space	Schwarzschild, polar $\ell = 2$
Polar reach and current?	two physical ingoing amplitudes modulo conformal gauge; Einstein self-block null; nonzero mixed fixtures	conformal direction an exact off-shell degeneracy
Polar disposition?	Einstein characteristics, all-decaying Coulomb asymptotics; endpoint nonselection parity complete	real frequencies, $\ell = 2$
Thermal weighting?	monodromy factor $e^{-\omega/T_H}$ universal across branches and parities	REDUCED-MODE; no quantum state
Stable/ringing?	not established 22	complex frequencies open; no QNM

13 What is not proved

The paper does not establish:

- a complete exterior initial-boundary well-posedness theorem: the real-frequency incoming connection is now exact and invertible on the entire positive real axis, but the outgoing reflection block and the complex-frequency connection remain unclassified;
- convergence or summability of the generic- ℓ formal series, axial X_2 , the deeper polar filtration on the $Q_{21} = 0$ wall, the terminal-only polar prefixes, or a non-Einstein-background Ricci-carrier composition;
- a canonical splitting of the Bach solution space, or complete metric reconstruction from every Ricci-carrier solution;
- a representative-invariant sign theory for the carrier self-pairing (the null-quotient pairing);
- time-domain boundedness, decay, completeness, quasinormal modes, or ringdown waveforms: boundary dévissage excludes growing separated modes of the complete non-split axial $\ell = 2$ system, but does not prove a full PDE stability theorem; the damped-half-plane frame events $\{i/4, i/2, i\}$ are not classified as Evans zeros or nonzeros;
- a second-order physical-process first law or nonlinear horizon dynamics;
- asymptotically flat conformal-gravity scattering with a complete boundary phase space, metric/tetrad falloffs, gauge-audited charge algebra, and horizon-to-infinity connection map: the scoped endpoint L^2 trace spaces and Grams do not by themselves supply these global structures;
- Hawking states, particles, grey-body factors, luminosity, a quantum BRST quotient, positivity, or unitarity: the monodromy temperature is a mode-level continuation factor, not a quantum state construction.

In particular, the paper withdraws the broad phrase “no local causal truncation.” Zero Cauchy data for the Ricci carrier is a local linear truncation. What is established is failure of the tested endpoint diagnostics to imply that truncation.

14 Certificate and reproduction ledger

The manuscript is an OUTLINE-ALLOWED/DRAFT-ALLOWED integration of committed certificates at repository baseline recorded in its generated claim map.

Claim	Authoritative certificate
Static spherical family	black_hole_programme/certificates/BH0_STATIC_SPHERICAL_BACKGROUND.json
Normalized generator and first law	black_hole_programme/certificates/BH1A_NORMALIZED_GENERATOR.json
Linear spherical dynamical extension	black_hole_programme/certificates/BH1B_DYNAMICAL_EXTENSION.json
Axial operator and Ricci–Bach composition	black_hole_programme/certificates/BH2A_AXIAL_OPERATOR.json
Extra horizon reach	black_hole_programme/certificates/BH2A_HORIZON_REACH.json
Action-derived axial current	black_hole_programme/certificates/BH2A_FLUX_MATRIX.json
Mixed and extra horizon flux fixtures	black_hole_programme/certificates/BH2A_CROSS_FLUX.json
Axial causal disposition	black_hole_programme/certificates/BH2A_CAUSAL_DISPOSITION.json
Polar trace-coupled split	black_hole_programme/certificates/BH2B_POLAR_SPLIT.json
Polar carrier horizon reach (conformal gauge quotient)	black_hole_programme/certificates/BH2B_POLAR_REACH.json
Polar Einstein two-dimensional reduction	black_hole_programme/certificates/BH2B_POLAR_EINSTEIN.json
Polar Einstein self-block null; conformal degeneracy	black_hole_programme/certificates/BH2B_POLAR_FLUX.json
Polar composition and mixed flux fixtures	black_hole_programme/certificates/BH2B_POLAR_CROSS_FLUX.json
Polar causal disposition	black_hole_programme/certificates/BH2B_POLAR_DISPOSITION.json
Horizon monodromy temperature	black_hole_programme/certificates/BH4_HAWKING_MONODROMY.json
Static-sector log classification	black_hole_programme/certificates/BH2_OMEGA_ZERO.json
Generic- ℓ axial Ricci-carrier recurrences (complete metric and current dispositions superseded)	black_hole_programme/phase2/general_l_axial_asymptotics/certificate.json
Complete axial six-dimensional module, repaired X_0 , and legacy E_0 supersession	black_hole_programme/phase3/axial_complete_reconstruction_repair/certificate.json; reports/phase3-black-hole-axial-complete-reconstruction-repair-2026-07-22.md
Exact axial null-endpoint wave-packet traces and action-derived flux Grams	black_hole_programme/phase3/axial_wavepacket_null_trace/certificate.json; black_hole_programme/phase3/axial_null_flux_gram/certificate.json; reports/phase3-black-hole-axial-null-flux-gram-2026-07-23.md
Axial RW/RW/spin-one filtration and invertible incoming connection	black_hole_programme/phase3/axial_rw_lx_triangular_preflight/certificate.json; black_hole_programme/phase3/axial_incoming_connection_analytic/certificate.json; black_hole_programme/phase3/axial_incoming_extended_domain_audit/certificate.json; reports/phase3-axial-incoming-extended-domain-audit-2026-07-23.md; black_hole_programme/phase3/axial_boundary_devissage_no_growth/certificate.json; reports/phase3-axial-boundary-devissage-no-growth-2026-07-23.md; black_hole_programme/phase3/axial_horizon_grassmann_mobius_to_r4_taylor2/future_horizon_outward_gram.json; black_hole_programme/phase3/axial_horizon_grassmann_mobius_to_r4_taylor2/future_horizon_factor_quotient.json; black_hole_programme/phase3/axial_spin_two_scattering_extension_preflight/certificate.json; black_hole_programme/phase3/axial_qnm_local_smith_dichotomy/certificate.json;

Every certificate records a schema, exact or controlled-numerical payload, independent verifier, mutations or controls, command receipt, dependency tags, and missing claims. The former axial and polar infinity-flux certificates, the infinity-selection half of the endpoint assembly, and the additional-log-tail half of the exterior boundary-value gate are historical inputs, not active authorities for infinity selection. Their unaffected Einstein, horizon, and leading-symbol components are represented above by independent certificates. The exact superseded identifiers are recorded in the machine-readable claim map. Exact axial endpoint trace spaces and flux Grams now combine with the invertible incoming connection to give globally populated incoming channels. A complete asymptotically flat scattering theorem still requires the outgoing/horizon map, regular-tetrad falloff, differentiable phase space, charge algebra, and complex-frequency audit.

15 Outlook

The formal infinity problem now has a different shape from the one suggested by the original fixture. The complete axial $\ell = 2$ pilot contains a finite repaired non-Einstein direction, but that finite class is stable only under the rate-zero Einstein shear and not under the true oscillatory shear. The polar module separately contains a generic-angular, generically nonradical mixed finite line, with a finite algebraic set of exceptional frequencies at each harmonic. These facts neither establish a representative-independent axial quotient nor restore an Einstein-selection principle.

The first global map is now exact:

$$\text{future-horizon-regular data} \xrightarrow[\cong]{T_-} \text{incoming } \mathcal{J}^- \text{ trace.}$$

The remaining decisive map is the outgoing/horizon scattering relation. It must include a regular tetrad or gauge, a gauge-audited Lee–Wald charge algebra, and the connection amplitudes at both future boundaries. A first validated full-amplitude attempt certified its diagonal carrier and kernel ranks and all local lower-lift solves on the first frequency cell, but ended in **SHORTFALL**: fixed interval frames discarded the shared ω -dependence needed for cumulative lower-block cancellation. The analytic triangular proof bypasses that limitation for T_- , not for T_+ . Complex-frequency work is likewise limited to the existing meromorphic axial continuation; no Wronskian zeros, quasinormal spectrum, or stability theorem has been computed.

The present result already fixes an important baseline:

Additional curvature directions occur in the future-horizon carrier, and independently the formal infinity modules contain finite non-Einstein or mixed directions. Neither leading endpoint diagnostics nor formal radial pairing finiteness selects the Einstein kernel. More strongly, the complete three-dimensional indefinite incoming trace is globally populated by future-horizon-regular solutions at every real positive frequency. The outgoing reflection block, complex-frequency poles, stability, CPT positivity, and a complete scattering operator remain open.

A Explicit axial endpoint Grams and uniform bounds

In the ordered bases

$$\mathcal{J}^- : (\text{XI0}, \text{XI1}, \text{EI0}), \quad \mathcal{J}^+ : (\text{XI2}, \text{XI3}, \text{EI2}),$$

the normalized incoming past Gram is

$$G_- = \begin{pmatrix} \frac{576}{5\omega} & \frac{96i}{5} & \frac{384\omega}{5} \\ -\frac{96i}{5} & -\frac{144\omega}{5} & -\frac{192i\omega^2}{5} \\ \frac{384\omega}{5} & \frac{192i\omega^2}{5} & 0 \end{pmatrix}, \quad (43)$$

and the normalized outgoing future Gram is

$$G_+ = \begin{pmatrix} \frac{384(-512\omega^6 + 640\omega^4 - 278\omega^2 + 39)}{5\omega} & \frac{12288i\omega^4 - 3072\omega^3 - 9984i\omega^2}{5} + 192\omega + 384i & \frac{-1536i\omega^2 + 384\omega + 768i}{5} \\ \frac{-12288i\omega^4 - 3072\omega^3 + 9984i\omega^2}{5} + 192\omega - 384i & -\frac{768\omega^3}{5} + 48\omega & \frac{96\omega}{5} \\ \frac{1536i\omega^2 + 384\omega - 768i}{5} & \frac{96\omega}{5} & 0 \end{pmatrix}. \quad (44)$$

The full current Grams are $\pi\alpha_W G_\pm$.

The exact squared Frobenius functions have certified maxima

	$\max_{\omega \in [1/2, 3/4]} \ G_\pm\ _F^2$	$\max_{\omega \in [1/2, 3/4]} \ G_\pm^{-1}\ _F^2$	
G_-	1429056/25	7025/65536	
G_+	10389888/25	19825/65536.	

(45)

For G_- , both functions are strictly decreasing. For G_+ , exact Sturm counts find one interior critical point for each function, and the endpoint derivative signs are $(-, +)$, proving that the critical point is a minimum. These facts prove (36) without inferring a lower bound merely from the determinants.

References

- [1] R. Bach, *Zur Weylschen Relativitätstheorie und der Weylschen Erweiterung des Krümmungstensorbegriffs*, Math. Z. **9** (1921) 110–135.
- [2] R. J. Riegert, *Birkhoff's theorem in conformal gravity*, Phys. Rev. Lett. **53** (1984) 315–318.
- [3] P. D. Mannheim and D. Kazanas, *Exact vacuum solution to conformal Weyl gravity and galactic rotation curves*, Astrophys. J. **342** (1989) 635–638.
- [4] T. Regge and J. A. Wheeler, *Stability of a Schwarzschild singularity*, Phys. Rev. **108** (1957) 1063–1069.
- [5] F. J. Zerilli, *Effective potential for even-parity Regge–Wheeler gravitational perturbation equations*, Phys. Rev. Lett. **24** (1970) 737–738.
- [6] J. Lee and R. M. Wald, *Local symmetries and constraints*, J. Math. Phys. **31** (1990) 725–743.
- [7] V. Iyer and R. M. Wald, *Some properties of Noether charge and a proposal for dynamical black hole entropy*, Phys. Rev. D **50** (1994) 846–864, [arXiv:gr-qc/9403028](#).
- [8] J. Maldacena, *Einstein gravity from conformal gravity*, [arXiv:1105.5632](#).
- [9] H. Lü, Y. Pang, and C. N. Pope, *Conformal gravity and extensions of critical gravity*, [arXiv:1106.4657](#).

- [10] S. Deser, E. Joung, and A. Waldron, *Partial masslessness and conformal gravity*, [arXiv:1208.1307](#).
- [11] H. Liu, H. Lü, and M. Luo, *On black hole stability in critical gravities*, [arXiv:1104.2623](#).
- [12] Y.-X. Chen, H. Lü, and K.-N. Shao, *Linearized modes in extended and critical gravities*, [arXiv:1108.5184](#).
- [13] E. A. Bergshoeff, O. Hohm, J. Rosseel, and P. K. Townsend, *Modes of log gravity*, [arXiv:1102.4091](#).
- [14] H. Lü, Y. Pang, C. N. Pope, and J. F. Vázquez-Poritz, *AdS and Lifshitz black holes in conformal and Einstein–Weyl gravities*, [arXiv:1204.1062](#).
- [15] Y. S. Myung, *Stability of Schwarzschild black holes in fourth-order gravity revisited*, [arXiv:1306.3725](#).
- [16] M. Momennia and S. H. Hendi, *Quasinormal modes of black holes in Weyl gravity: electromagnetic and gravitational perturbations*, [arXiv:1910.00428](#).
- [17] C.-Y. Chen and P. Chen, *Gravitational perturbations of non-singular black holes in conformal gravity*, [arXiv:1902.01678](#).
- [18] M. Momennia, S. H. Hendi, and F. S. Bidgoli, *Stability and quasinormal modes of black holes in conformal Weyl gravity*, [arXiv:1807.01792](#).
- [19] G. Antoniou, L. Gualtieri, and P. Pani, *Gravitational quasinormal modes in quadratic gravity*, Phys. Rev. D **111** (2025) 064059.
- [20] R. A. Konoplya, A. Spina, and A. Zhidenko, *Time evolution of black hole perturbations in quadratic gravity*, [arXiv:2505.01128](#).
- [21] D. Langlois, K. Noui, and H. Roussille, *Black hole perturbations in modified gravity*, [arXiv:2103.14750](#).
- [22] P.-K. Hung, J. Keller, and M.-T. Wang, *Linear stability of Schwarzschild spacetime: decay of metric coefficients*, [arXiv:1702.02843](#).
- [23] H. Masaoood, *A scattering theory for linearised gravity on the exterior of the Schwarzschild black hole II: the full system*, [arXiv:2211.07462](#).
- [24] T. Adamo, S. Nakach, and A. A. Tseytlin, *Scattering of conformal higher spin fields*, JHEP **07** (2018) 016, [arXiv:1805.00394](#).
- [25] K. S. Stelle, *Renormalization of higher-derivative quantum gravity*, Phys. Rev. D **16** (1977) 953–969.